This is well defined because if $\pi(a) = \pi(a')$, then

$$\pi(a'b) = \pi(ab + (a' - a)b) = \pi(ab) + \pi((a' - a)b) = \pi(ab)$$

because $(a' - a)b \in Ib \subseteq I = \ker \pi$, as I is a right ideal; similarly, if $\pi(b) = \pi(b')$, then

$$\pi(ab') = \pi(ab + a(b' - b)) = \pi(ab) + \pi(a(b' - b)) = \pi(ab)$$

because $a(b' - b) \in aI \subseteq I = \ker \pi$, as I is also a left ideal. Thus, A/I is an algebra.

Similarly, if V is a representation of A and $W \subset V$ is a subrepresentation, then V/W is also a representation. Indeed, let $\pi : V \to V/W$ be the quotient map, and set $\rho_{V/W}(a)\pi(x) := \pi(\rho_V(a)x)$.

Above we noted that left ideals of A are subrepresentations of the regular representation of A, and vice versa. Thus, if I is a left ideal in A, then A/I is a representation of A.

Problem 2.5.1. Let $A = k[x_1, \ldots, x_n]$ and let $I \neq A$ be any ideal in A containing all homogeneous polynomials of degree $\geq N$. Show that A/I is an indecomposable representation of A.

Problem 2.5.2. Let $V \neq 0$ be a representation of A. We say that a vector $v \in V$ is **cyclic** if it generates V, i.e., $Av = V$. A representation admitting a cyclic vector is said to be **cyclic**. Show the following:

(a) V is irreducible if and only if all nonzero vectors of V are cyclic.

(b) V is cyclic if and only if it is isomorphic to A/I, where I is a left ideal in A.

(c) Give an example of an indecomposable representation which is not cyclic.

Hint: Let $A = \mathbb{C}[x, y]/I_2$, where I_2 is the ideal spanned by homogeneous polynomials of degree ≥ 2 (so A has a basis $1, x, y$). Let $V = A^*$ be the space of linear functionals on A, with the action of A given by $(\rho(a)f)(b) = f(ba)$. Show that V provides such an example.